

Unit ID: 679b1051aaf593b2823f31a5

Unit 3 - Programming a Countdown

PDF Documents

PDF Document 1: 1738215503432-Chapter 7 - Unit 3 - Programming a Countdown.pdf

This PDF document is included in the unit materials.

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UNIT 3 PROGRAMMING A COUNTDOWN

What is a Countdown?



Imagine you're about to open a present on your birthday. You might count down from 5: "5, 4, 3, 2, 1!" and then open the present. This is a countdown.

A countdown is when we count backwards from a number to zero to mark the start of something. It's like a timer that helps us know when something is about to happen.

HOW IT WORKS IN ACTION:

We're going to use something called MODI to make our own countdown. MODI is like a set of electronic building blocks that let us create fun projects.

WHAT WILL WE DO?

PRESS A BUTTON

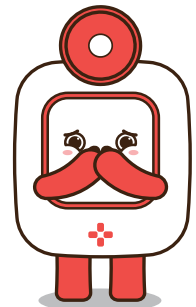
This will start our countdown.

WATCH THE COUNTDOWN

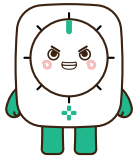
The MODI will show us the numbers counting down from 5 to 1.

HEAR THE COUNTDOWN

The MODI might make a sound to let us know when the countdown is over.



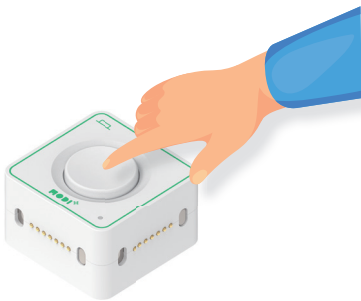
Do you remember which action Button can received?



Do you remember which action Button can received?

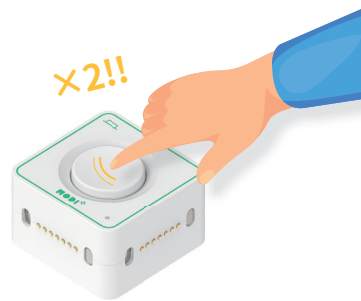
Button can receive 4 types of information.

1. CLICK



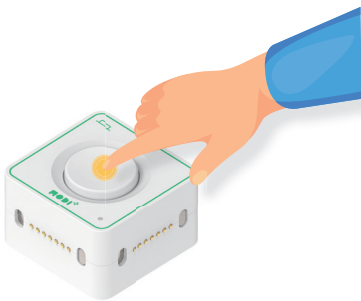
Detects if Button is clicked or not.

2. DOUBLE CLICK



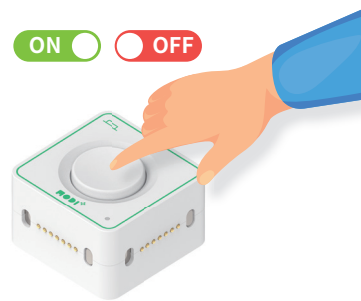
Detects if Button is double - clicked or not.

3. PRESS



Show the press status of Button.

4. TOGGLE



Changes on / off when Button is clicked.



Activity: Create **Your Own Smart Wastebin**

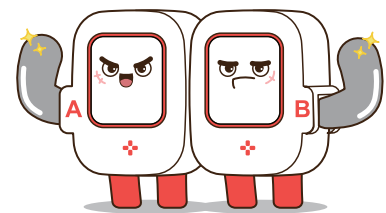
1 Materials Needed:



Button



Speaker



2 Code story

Turn off Speaker



If the Button is clicked



Wait 5 seconds

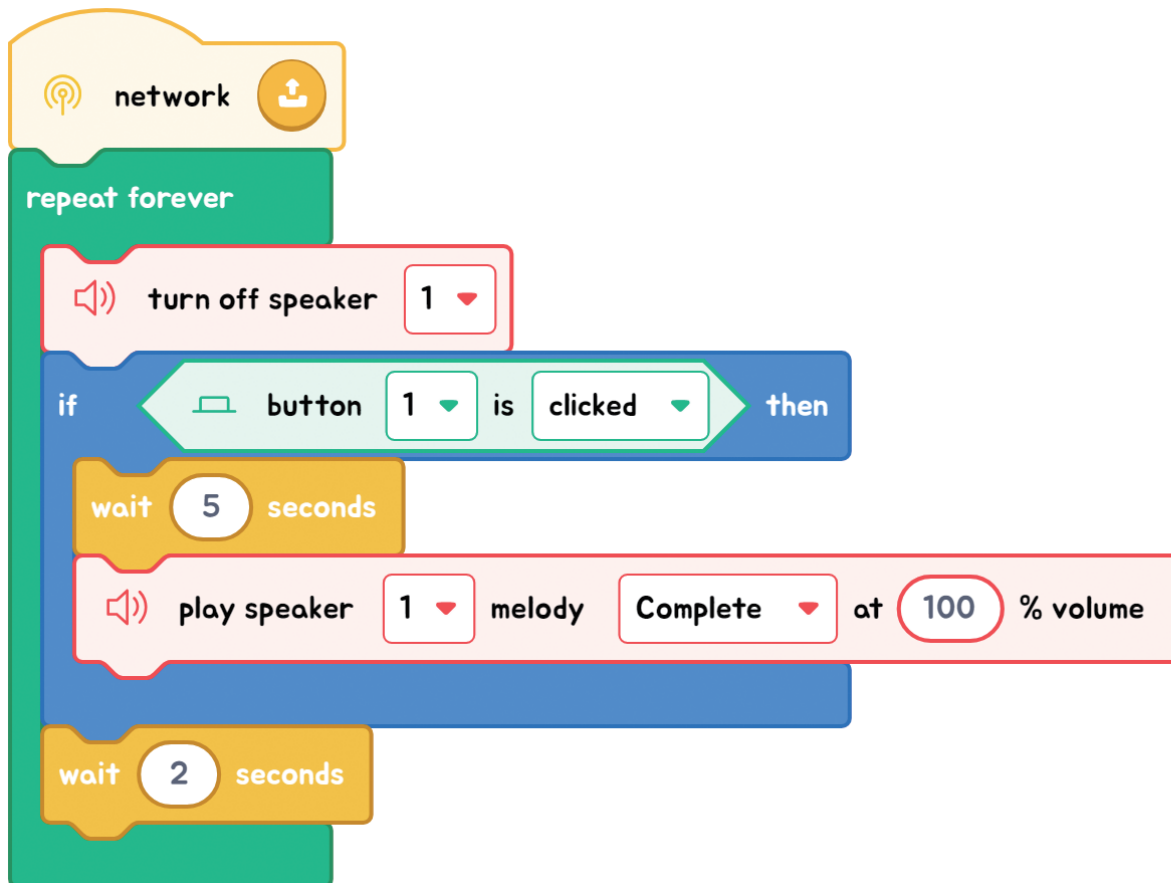


Play complete sound on the Speaker



Wait 2 seconds

Code Quest



Checkpoint

Let's add a code that shows a red light when the countdown is done.

HINT!

When the button is not clicked, the LED has to be off.

